

METROPOLIS, IL, USA

WMO: 720170

Lat: 37.186N

Lon: 88.751W

Elev: 384

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63851

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	16.2	-1.5	5.2	15.4	4.4	7.0	19.5	21.2	43.7	19.5	42.9	3.9	10	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	93.3	76.1	91.2	75.9	89.9	75.2	80.1	88.4	78.5	86.8	77.4	85.8	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.6	145.7	84.6	76.5	140.2	83.2	75.0	133.1	81.5	43.9	88.3	42.3	87.3	41.0	85.1	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.9	12.9	2.0	96.4	9.7	3.2	-4.9	98.7	-10.6	100.6	-16.0	102.4	-23.0	104.7		
(5)			1.9	82.4	9.1	2.9	-4.6	84.5	-10.0	86.2	-15.1	87.8	-21.7	89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	35.5	38.8	48.3	58.4	67.8	75.6	78.3	76.4	70.5	59.0	47.1	39.4
	DBStd	17.52	11.19	11.38	10.71	9.54	7.76	8.21	4.83	4.94	6.40	8.86	9.25	9.85
	HDD50	1552	468	345	163	33	0	10	0	0	0	19	159	355
	HDD65	4089	916	735	524	234	66	24	0	1	22	232	539	796
	CDD50	4480	17	30	112	285	551	779	877	819	616	297	72	25
	CDD65	1540	0	0	8	35	152	342	412	355	188	44	3	1
	CDH74	15046	0	0	54	316	1385	3492	4200	3352	1791	444	12	0
	CDH80	5909	0	0	3	41	424	1448	1825	1385	668	114	1	0
(14) Wind	WSAvg	4.7	6.2	6.0	6.3	6.1	4.6	3.8	3.1	2.6	3.0	4.0	4.8	5.5
(15) Precipitation	PrecAvg	49.5	4.2	3.6	4.9	4.9	5.0	4.1	4.2	3.1	3.3	3.7	4.0	4.4
	PrecMax	72.8	7.8	8.8	11.1	16.0	11.0	10.2	9.9	9.2	10.2	10.7	9.8	9.3
	PrecMin	29.1	1.2	1.2	1.9	0.4	0.4	0.8	1.4	0.4	0.0	0.1	0.7	1.3
	PrecStd	9.5	1.6	1.7	2.2	2.7	2.4	2.2	2.4	1.9	2.3	2.4	2.3	2.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	71.2	79.5	84.0	90.3	95.3	98.2	97.1	93.1	88.1	75.1	68.9
		MCWB	61.6	61.4	64.2	67.9	72.0	73.7	76.5	75.7	74.6	72.5	64.1	64.1
	2%	DB	62.8	67.4	74.4	81.1	87.4	91.8	93.1	92.5	90.5	83.0	71.7	64.1
		MCWB	58.5	59.3	62.1	64.8	71.1	75.2	78.3	74.9	74.0	69.3	61.6	59.8
	5%	DB	58.9	62.5	70.4	77.9	85.0	90.4	91.0	90.3	87.7	79.2	67.4	60.0
		MCWB	54.0	55.6	60.0	63.5	69.9	74.5	77.6	75.2	73.1	66.3	58.5	55.1
	10%	DB	54.1	57.3	65.9	74.5	81.9	88.1	89.2	87.8	83.8	74.7	63.2	55.2
		MCWB	49.3	51.8	56.9	62.4	68.4	73.8	76.6	74.7	71.0	64.5	55.5	51.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.1	64.3	66.8	71.3	75.4	81.0	82.1	81.7	77.6	74.6	68.2	64.5
		MCDB	64.3	69.2	75.1	78.7	84.0	89.4	91.3	88.4	87.7	84.1	72.4	67.2
	2%	WB	60.0	60.5	64.0	68.1	73.8	78.6	80.5	79.1	76.3	72.0	64.0	61.0
		MCDB	62.0	64.8	71.7	76.7	82.8	86.9	89.2	87.3	86.0	79.2	68.2	63.8
	5%	WB	54.9	57.2	61.6	66.1	72.3	77.2	79.2	77.7	75.0	69.5	60.1	56.6
		MCDB	58.1	61.5	68.8	74.2	81.0	86.0	87.6	86.1	84.0	75.4	65.0	59.4
	10%	WB	49.8	52.3	59.0	64.3	70.7	75.7	77.8	76.5	73.3	66.7	56.4	51.6
		MCDB	53.5	56.6	64.9	71.6	78.6	84.4	85.9	84.5	80.7	72.8	61.5	54.8
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.0	19.3	22.0	20.1	20.8	18.9	20.2	22.4	22.4	19.9	15.7
		MCDBR	21.1	23.0	24.6	25.1	23.4	21.8	20.5	22.9	25.4	26.8	25.2	20.5
		MCWBR	18.1	17.4	14.8	12.6	9.5	7.8	7.9	8.6	10.1	12.7	16.2	17.1
	5% WB	MCDBR	18.4	20.1	21.1	20.1	19.6	18.8	18.4	19.0	22.1	21.6	20.0	17.9
		MCWBR	17.7	16.9	13.7	12.0	9.1	8.4	8.2	8.7	10.1	11.6	16.5	16.9
(40) Clear-Sky Solar Irradiance	taub		0.319	0.329	0.365	0.400	0.428	0.440	0.479	0.478	0.411	0.361	0.324	0.313
	taud		2.459	2.413	2.370	2.274	2.250	2.228	2.145	2.136	2.321	2.444	2.491	2.481
	Ebn at Noon		275	287	285	279	272	268	256	254	266	272	272	270
	Edh at Noon		27	31	36	42	43	45	48	48	37	30	25	24
(44) All-Sky Solar Radiation	RadAvg		677	874	1212	1595	1799	2027	1954	1793	1501	1134	775	577
	RadStd		65	107	106	101	154	127	122	93	92	123	83	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page